

Research Volunteer Information Sheet

A Comparative Multiparametric MRI Evaluation of Morphological Sequences, T2 Mapping, and Diffusion Imaging for the Characterisation of Healthy Knee Cartilage.

You are being invited to participate in a research study. Before you decide whether or not you would like to take part, it is important that you read this document **carefully**. Here you will find information about the study, what it will involve, the best way to prepare for it and how we shall be using your data. It also explains your legal rights concerning the data we collect from you. If anything is not clear, or if you would like more information, do feel free to ask us. You may find our contact details at the end of the document.

What is the purpose of this study?

This study aims to assess the diagnostic value of a multiparametric MRI framework that integrates morphological imaging, T2 mapping, and diffusion-based techniques for the characterisation of healthy knee cartilage. Conventional MRI methods, such as T1-weighted and proton density imaging, are primarily limited to detecting structural abnormalities and cannot identify subtle biochemical or microstructural changes that precede visible degeneration. By incorporating quantitative MRI parameters including T2 relaxation times, apparent diffusion coefficient (ADC), and fractional anisotropy (FA), this research seeks to provide a more comprehensive evaluation of cartilage composition and organisation.

The study objectives are to:

- Develop and standardise MRI acquisition and segmentation protocols.
- Establish normative reference values for healthy knee cartilage.
- Evaluate the sensitivity of quantitative MRI biomarkers to biomechanical loading.
- Determine whether multiparametric maps improve diagnostic confidence compared with conventional imaging alone.

The research will adopt a quantitative experimental approach using a 3T MRI system at the University of Malta. Healthy volunteers aged 18 to 30 years will undergo MRI scanning using proton density fast spin echo sequences for morphology, T2 mapping, and diffusion-based imaging for microstructural assessment. Manual segmentation will be used to delineate the regions of interest within the articular cartilage, particularly in weight-bearing zones. Finally, quantitative parameters will be extracted to establish baseline reference data, and radiologists will evaluate the added diagnostic value of multiparametric maps.

This study is expected to enhance the early detection of cartilage abnormalities, contribute to the standardisation of quantitative MRI methods in clinical practice, and support the development of reproducible imaging biomarkers for future diagnostic and research applications.

Who is organising this study?

This study is organised by the University of Malta. It has been given ethical clearance by the University of Malta's Research Ethics Committee (UREC).

The **principal investigator (the researcher in charge of the study)** is Claude Bajada.

What will I be expected to do if I take part?

Volunteers will undergo a 15-minute MRI scan of the knee. Prior to imaging, each participant's height, weight, and Knee injury and Osteoarthritis Outcome Score (KOOS) questionnaire will be recorded.

How should I prepare for the scan?

MRI is considered a very safe technology, but it can be more risky if you have certain medical implants or metal fragments in your body. When you arrive at the scanning centre, therefore, a member of staff will ask you a few questions to check that it is safe for you to participate. As a natural precaution, we do not take scans of pregnant women (although there is currently no evidence of danger). We kindly ask you not to take part if you are pregnant or have any reason to think you may be pregnant.

You will need to remove any metal accessories before entering the MRI room, due to the strong magnetic field of the scanner. Objects made of metal or that can be damaged by the magnetic field (such as keys, watches, coins, credit cards, mobile phones and hearing aids) will have to be left in the ante-room as well. It is best to avoid wearing sports clothing, as this may contain metal fibres, but in any case hospital gowns are provided and if the staff in charge thinks it is safer for you, they will ask you to change. You will also be instructed to use the changing room to remove any underwear with metal fastenings or underwiring. **Please do not wear any makeup, talc, hair products, cosmetic contact lenses (these are lenses which change the way your eyes look) or prosthetic contact lenses** when you come for your scan.

The scanning session will take 15 minutes, and it is important that you remain as still as possible while the scanner is operating.

What is MRI?

The term 'Magnetic Resonance Imaging (MRI)' refers to a technique which makes use of radio waves and strong magnets to produce high-quality images of different parts of the body. The radio waves are considered to be safe for humans (they are similar to those which carry radio and TV signals). No drugs, injections, and X-rays (or any other ionising radiation) will be involved.

The images the MRI scanner will take of your knee will consist of:

- Morphological MRI - these images provide high quality reference of your knee and cartilage structure
- Diffusion MRI - these images give us information about the microstructure of the Knee.

Are there any risks or side effects I should be aware of?

The noise produced by the scanner can be loud, so we will provide you with earplugs or headphones (these do not eliminate the noise completely, but reduce it significantly). Please note that the inside of the scanner is quite small. If you are claustrophobic (that is, if you cannot tolerate confined spaces), it may be better for you not to take part.

MRI is non-invasive and considered safe. It does not have any known risks or side effects, except that in a small number of people, the scanning can cause small twitches or a tingling sensation in peripheral parts of the body. Being in the magnetic field may also, in very rare cases, bring on vertigo (dizziness).

You will have a buzzer in your hand while in the scanner. Should you experience any of the abovementioned effects, or if you feel uncomfortable at any point during the scanning, please alert us by pressing the buzzer. We will then speak to you via an intercom and if necessary will stop the scan. Remember that you may opt out of the study at any time without being obliged to explain why.

Data will be transferred from the MRI scanner to UM computers where it will be analysed. In any data transfer there is a minimal risk of a data breach associated with data transfer and storage. To mitigate the risk of a data breach, data will only be transferred using current security standards.

What are the possible benefits of taking part?

There will not be any direct benefit to you. However, the research for which your data is utilised may help improve diagnosis and treatment of diseases.

It is possible that some of the research conducted using your data could lead to the development of new diagnostic tests, drugs, technology or other commercial products. If this happens, please note that you will not receive any portion of the profits generated from these products, nor will you have any ownership rights over them.

Will my images be looked at by a doctor?

This scan is NOT meant to be diagnostic. We do have a review system whereby the MRI images of study participants are looked at by a professional. However, research scans are often not of the same high quality as clinical ones, nor is the quantity of images collected always sufficient to reveal any irregularities that might be present. So if an abnormality exists - even if it is potentially serious - THERE IS NO GUARANTEE that it will be picked up from your scans. You should always speak to your doctor if you have any health concerns.

What is an incidental finding?

There is a small chance that undergoing an MRI scan results in an incidental finding i.e. the discovery of an abnormality that you are not currently aware of.

Should an irregularity be detected in your MRI images, a specialist doctor (radiologist) will decide whether you are to be informed. If the abnormality is unlikely to be relevant for your health, it will not be disclosed to you. In all other cases, you will receive an email from the University of Malta MRI Platform (UMRI) that advises you to contact a doctor you trust, who would be able to refer you for a full clinical scan. In addition, UMRI will decide based on advice from the radiologist whether to contact you by phone or not. Please note that even if a serious abnormality indicates a disease or condition for which no treatment or surgical procedure is readily available, we would still be letting you know about it should we notice it. Please note that the majority of anomalies that can be revealed by MRI scans are not a cause for concern.

We want you to be aware that if a significant abnormality is detected, it could lead to an early diagnosis and start of treatment, which might be beneficial to you. However, incidental findings can sometimes result in uncalled-for anxiety and unnecessary medical tests/treatments, especially if the abnormality is of uncertain clinical relevance, if it later turns out to be benign, or if it reveals an illness or disorder for which there is no known treatment. In a small number of cases, unanticipated findings

may affect your prospects for future employment and make it more difficult to obtain certain kinds of insurance.

What if an incidental finding is made over a year after data collection?

In these circumstances there would be no way to trace the scans to you, so if an abnormality is detected it would not be possible for us to let you know about it.

What happens at the end of the study?

The results may also be used for public outreach and/or teaching purposes, presented at scientific meetings, and written about in journals and/or dissertations. This will always be done in a manner which makes it impossible to identify individual volunteers.

Will I receive financial compensation?

No

Is it possible to withdraw from the study?

Yes. Your participation is totally voluntary, and you may opt out at any time, without being asked to explain why. Nor will there be any repercussions on your medical care or legal rights. The same holds true if you decline to take part.

Kindly note that any data collected before your withdrawal could still be stored and used as outlined in this document unless you explicitly ask us to erase it or restrict its processing (refer to the question '*Can I object to my data being processed by the University of Malta?*' for more information. Please be aware that we might not always be able to fully comply with erasure and/or restriction of processing requests, although we will try our utmost.)

Can the researchers interrupt the study?

Although this is unlikely, the research may be discontinued at any time. The researchers would not be able to use your data for the study should this happen. Additionally, some unforeseen circumstances (for instance, a lot of movement during the scanning) can compromise the quality of the scans and prevent the researchers from including your data in the study.

In any of these situations, your data could still be stored and utilised as described in this document.

How will my identity be protected?

Your data will be assigned a unique identification code, and your name will not appear on any research data or questionnaires. **Once all the data needed for the project has been collected, only the principal investigator** (the main researcher responsible for the study) **will retain information that links the code to your name** (this information is essential because without it we would not be able to identify you in case of an incidental finding), and he/she will not pass it to anyone, except to the designated UMRI personnel in the eventuality that an abnormality is detected or in the event that you lodge a request with UMRI for a copy of your MRI data. **The principal investigator will be asked to keep said information secure and confidential (and separate from your scans/any questionnaires you fill in), on encrypted, offline storage devices accessible only to the principal investigator, and to dispose of it one year after the last day of data collection.**

What will be done with the data?

A copy of your data will be kept securely in a storage system (repository) at the University of Malta. It will be released for use in this study, and potentially also for other research projects (including research into the development of imaging technology or biomedical engineering applications) and teaching. A lot of effort has been put into ensuring that the data transmission pipeline and storage servers are secure and protected against unauthorised access.

Confidentiality - who will have access to the data?

Original data on the scanner:

The data collected from you will NOT be stored in repositories for patient data at Mater Dei Hospital. A copy will be transferred to the University repository, as explained above, while the original scans will remain on the MRI machine for about 4 weeks before being automatically deleted – the exact duration depends on how many people get an MRI and the amount of data acquired. During this time, physicists, Magnetic Resonance radiographers and the system engineer will have access to the data.

The imaging software requires the radiographer to input the date of birth, age, sex, height and weight of the person to be scanned. The data on the MRI machine (and the copy transmitted to the University) will consequently incorporate these attributes, together with other details specific to your scan, but NOT your name or national ID card number.

Copy of data in the University's repository:

The data in the main repository will be accessible to a small number of University of Malta employees who administer the system. This data will have been cleaned of all information that is unique to you or your scan, except for *useful attributes* like age, sex, weight, height and scan date (so called because they can be useful for research). The employees may process the data for administrative purposes (for example, to get it ready for research or teaching, or in case you want to exercise any of your GDPR rights).

Data released for research and teaching:

The repository data will be made available to: 1) the research team working on this particular study, and possibly 2) other researchers and students at the University of Malta involved in approved projects (please note that examiners assessing student projects might need to view the data, usually for verification purposes); 3) teaching personnel at the University for teaching activities (and students attending their courses); 4) authorised external entities (such as laboratories, the scanner manufacturer, or researchers not affiliated with the University) for research projects - in this case, any sharing of data would be subject to a data sharing agreement incorporating the safeguards that apply to the data kept in the University repository.

A full list of projects and principal investigators can be found on the UMRI platform website: <https://www.um.edu.mt/platforms/umri/>. The *useful attributes* we forward to researchers/teaching personnel will be limited to the ones that are strictly necessary, and will never include your name or national ID card number.

Any application for doing research with your data will have to be approved by a recognised ethics committee and given prior authorisation by the information and data protection commissioner (IDPC), as required by Maltese law.

Data sharing may take place via a secondary research repository holding copies of the data in the main repository that has been released for research. Researchers/teaching personnel would be given access to the research repository via user accounts that restrict the access to the data that they need.

What is the legal basis for processing my data for research and teaching purposes?

Personal data shall be processed in line with the General Data Protection Regulation (GDPR - EU Regulation 2016/679), the Data Protection Act (Cap. 586 of the Laws of Malta) and any other applicable legislation.

The GDPR requires that all data processing has a legal basis. The legal bases we will adopt to process your personal data for research and teaching objectives are:

Research:

- a) GDPR Article 6(1)(e) - the processing is necessary for the performance of a task carried out in the public interest;
- b) GDPR Article 9(2)(j) - the processing is necessary for archiving purposes in the public interest, scientific or historical research purposes.

Teaching:

- c) GDPR Articles 6(1)(a) and 9(2)(a) - we will only utilise your data for teaching if you explicitly consent to our doing so.

What categories of personal data are concerned?

General personal data (name, age, etc.)

Health and biometric data, which are considered special categories of personal data in the GDPR.

Who is the data controller for the research data?

The GDPR determines that a 'data controller' is the person or entity that, alone or jointly with others, determines the purposes and means of the processing of personal data.

The following is a description of the stages of processing and the associated controller/s:

- *Data collection.* Controllers - Mater Dei Hospital and the University of Malta
- *Data curation for scientific research* - University of Malta
- *Individual research studies* - University of Malta and/or the researchers concerned

If you have any queries about a particular aspect of the data processing, kindly contact the data controller who is responsible for processing that aspect of your data. In case there is more than one controller involved, you are free to get in touch with either party. Contact details are provided at the end of this document.

Please note that questions for any researchers conducting the study should be directed to the principal investigator.

How long will you store my data?

The University of Malta will store your data for a period of 10 years from the date of last use. If, after 10 years, the data is no longer deemed to be necessary for scientific research purposes, it will be erased. Note that data which has been approved for use in certain projects may be kept longer.

Before data is passed to researchers for a particular study, they will be required to fill in a compliance form. This will bind them to erase the data (and any derivatives that are not in aggregate form) from their computers/personal storage systems once the project has been completed - unless the data is needed for a follow-up study, in which case a new application has to be submitted to UMRI, and the data cannot be further processed or shared until authorization has been obtained from the relevant bodies.

Can I object to my data being processed by the University of Malta?

Yes – under the GDPR and the Malta Data Protection Act 2018, you have the right to request, in relation to your personal data:

- 1) Access
- 2) Rectification of any inaccuracies
- 3) Erasure
- 4) Restriction of processing
- 5) A copy of your data (data portability)

At any time you may also object, on grounds relating to your particular situation, to processing of personal data concerning you.

Requests should be made to UMRI in the first instance by filling out the GDPR form on the UMRI GDPR webpage: <https://www.um.edu.mt/platforms/umri/gdprrights/>.

Please note that the Malta Data Protection Act and the GDPR allow for derogations from the aforementioned rights (1 – 4 above) in circumstances where their exercise would seriously impede research objectives. For instance, this might be the case if your data has already been processed for a specific study and you ask for it to be erased.

We will always do our utmost to satisfy your requests. In situations where we are unable to do so, we will always explain the reason for this to you.

Where can I get more information?

Please feel free to ask us any questions you might have. More information may be obtained from:

- The University of Malta Magnetic Resonance Imaging Platform (UMRI), by sending an email at umri.platform@um.edu.mt or placing a call at [REDACTED]. Ideally this is your first contact point.
- The principal investigator, Claude Bajada, who can respond to queries related to the objectives and design of the study and may be contacted as by email at [REDACTED]

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- The Data Protection Officer of the University of Malta (dpo@um.edu.mt);
 - The Data Protection Officer of Mater Dei Hospital (dataprotect@mdh.gov.mt).

We recommend that you visit UMRI's [website](http://www.um.edu.mt/umri) [www.um.edu.mt/umri] if you want the most detailed and up-to-date information on data use and our current projects, but please do feel free to contact us if you need more details.

From time to time we may make some amendments to this information sheet. In such cases we will not notify you directly, but will make the latest version available on the website.

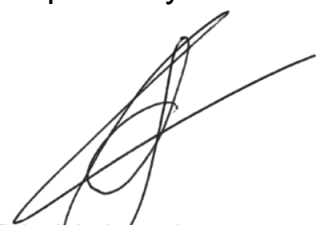
Can I lodge a complaint with an independent entity?

Yes. In the first instance, we recommend that you contact the UMRI Platform at umri.platform@um.edu.mt. You can also get in touch with the University of Malta's Data Protection Office by sending an email at dpo@um.edu.mt.

If you are not satisfied with the response, you may submit a complaint to the Office of Malta's Information and Data Protection Commissioner. There is a [dedicated online form](https://register.idpc.org.mt/report-breach/complaint/) [register.idpc.org.mt/report-breach/complaint/] that makes this process easy. Please visit the website www.idpc.org.mt/ for more information, or contact the office at idpc.info@idpc.org.mt or on +356 2328 7100.

Thank you for dedicating some of your time for this study! Our research would not be possible without the support and goodwill of people like you.

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